

FINOPTIMISE: A COMPREHENSIVE USER EXPERIENCE METHODOLOGY FOR ASSESSING A STOCK TRADING SIMULATION

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Abstract

Students undertaking a taught Masters Degree in Computer Science (in Technological University Dublin) are given the option for their capstone project of either completing an individual dissertation project or working on a group-based project. The group project provides the students with the opportunity to undertake a significant software development project from initial concept stage right through to delivery, in a team environment that closely represents the project environment in the IT industry. All students are expected to contribute to all aspects of the project, and they are all required to help devise and implement an UX evaluation process that is customized to their project.

This paper describes one such project that focused on the design, implementation and evaluation of a mobile-first web application called “FinOptimise”, that allows players to take part in a virtual stock trading game, and to help them build a healthy approach to stock investing. The FinOptimise system is designed to be a group-based game in which players compete against each other in a league, and each player has a virtual portfolio with an initial sum of virtual money, that they can invest in different stocks until the goals of that league are completed.

The User Experience methodology developed for this project can be categorised for convenience into five discreet stages (but in practice there was a degree of overlap and interaction between the stages):

- Phase 0: Market Research and Analysis
- Phase 1: Feature Prototyping and Design of Application
- Phase 2: Prototype End-to-End Usability Testing
- Phase 3: Autobiographical Testing
- Phase 4: System End-to-End Usability Testing

This methodology proved to be very valuable in helping to guide the design of the application towards a user-friendly and accessible system. It also demonstrated the importance of user feedback throughout the development process to ensure that the resultant system is both fun and functional.

Keywords: User Experience, UX Methodology, Stock Trading Game.

1 INTRODUCTION

This project was developed to help teach people about stock investment using a game-based learning approach. The application can, therefore, be considered a *serious game*, which is a game that is an “*entertaining tool with a purpose of education, where players cultivate their knowledge and practice their skills ... during gaming*” [7]. Zhonggen [16] notes that one of reasons for the effectiveness of serious games in education is their influence on learners’ moods, since serious games can be enjoyable, and therefore encourage players to continue the play, leading to increased learning. Anastasiadis, *et al.* [3] suggest that serious games can “*render learning of academic subjects more student-centered, engrossing, enjoyable, interesting and thus, more effective and efficient*”.

Serious games are part of a broader umbrella concept of Games-Based Learning, which looks at any combination of education and games. So, if serious games are ‘games with educational content’, then there is an antithetical concept called “gamification” which is ‘education with games content’ [1]. The system being developed in this project can be seen as existing on a spectrum of games-based learning, which started off as a simulation, and moved towards a “serious games” based on user feedback and expert feedback.

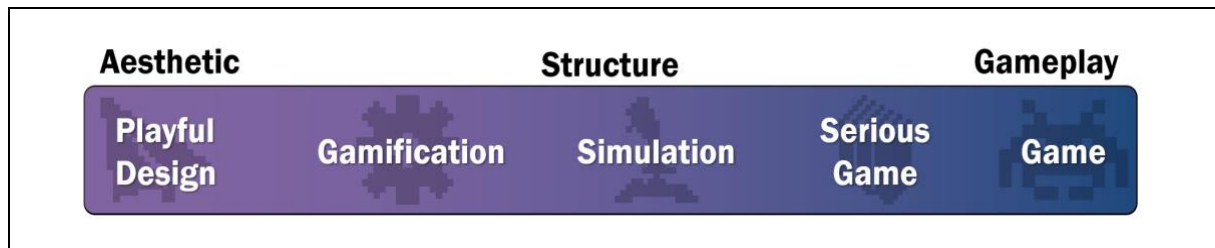


Figure 1. The Games-Based Learning Spectrum (Marczewski, 2016)

Some key factors identified as being important in the successful implementation of serious games were identified by Ravyse, *et al.* [13], and these include:

- **Game Narrative:** This factor is concerned with the storyline of the game that the players are expected to immerse themselves in.
- **Realism:** This factor is how closely the storyline matches the real world.
- **AI and Adaptivity:** This factor looks at whether there is any Artificial Intelligence (AI) used within the game, and does it allow the game to change in any way.
- **Interaction:** This factor means that games require user input and respond accordingly, in a usable and accessible way. The system should also communicate the current status of the game back to the user.
- **Feedback and Debrief:** This refers both to in-game feedback through a variety of in-game reward mechanics or interactions, as well as post-game debriefing and reflection sessions which ultimately elucidate the learning material and place the game-learning experience into a greater context.

Based on this survey of the literature, an outline of the design of the FinOptimise system is outlined in the following section.

2 DESIGN OF FINOPTIMISE

The design of the FinOptimise system underwent several iterations before the final format was reached. The concept of a virtual stock trading platform, which includes a gamified version of stock trading, was chosen primarily because a gap was identified with most existing stock trading platforms, whereby they tend to overload their users with information, including an excess of visual information that novice users are not familiar with. An initial survey as part of this research also indicated that many people would be interested in investing in stocks, had they the knowledge and necessary income, particularly if the system were similar to a Fantasy Football Leagues. Therefore a useful feature in the FinOptimise system is a recommender system, that suggests stock picks for the users based on their previous investment history, as well as the behaviour of other players in each particular league.

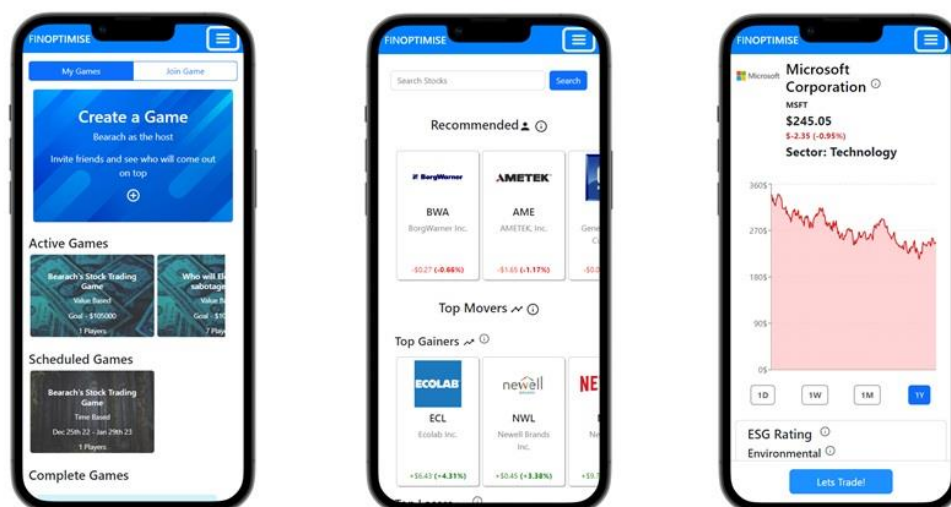


Fig. 2. Sample Screens of the FinOptimise system.

2.1 Front-End

Creating a positive user experience is essential to ensure users find it easy to start playing in a league, and also that they will return to the league on a regular basis. To help achieve this goal, the interface primarily follows the guidelines of Flat 2.0 design, which combines elements of skeuomorphic and flat design by adding some depth to mostly flat elements [5]. To explore potential interfaces with users, a number of prototypes were developed using the Figma prototyping application, and these were presented to users who gave feedback using the “think aloud” protocol [10].

2.2 Back-End

A crucial technical consideration is the hosting platform and the technical architecture of the FinOptimise system. Amazon Web Services (AWS) was chosen to host the application as it is cost effective and provides sufficient storage to record a wide volume of data on each organisation, including stock prices, as well as news and Twitter data. The basic technical architecture uses a MongoDB, Express JS, React and NodeJS (MERN) stack, for flexibility and ease-of-use.

2.3 Database

The database must store a wide range of information, including data about each user of the FinOptimise system, as well as historical data about each company and their performance, as well as all the key mediastream and social media data relevant to each company. MongoDB is a very good choice given how heterogeneous the data is, as it provides a great deal of flexibility in the type of data that can be stored, as well as how it is stored. It also has very powerful querying and analytics utilities built into it, and it can therefore be used to present complex content in a clear way.

2.4 Artificial Intelligence

The FinOptimise system uses machine learning both to calculate a sentiment score for media coverage of organisations, and for the recommender system that recommends stock picks. In terms of the sentiment score, each relevant newspaper article and Tweet will have a sentiment score (positive, neutral, and negative) associated with it, and the average sentiment from these is also displayed on each stock page. This provides information to the user in relation to how that company is doing in the media, which is often a good indicator of how the stock will perform [15]. In terms of the recommender system, typically such a system either recommends new content either on the basis of the history of the individual user (content-based), or it looks at users that the system deems to be sufficiently similar, and recommends new content based on their activities (collaborative-based).

3 EVALUATION OF FINOPTIMISE

The overall aim of this project is to provide a beginner friendly platform allowing users to experience virtual investing and to provide a starting point for building a healthy interest in investing. To achieve this, three main forms of evaluation were undertaken: Software Testing, Accessibility Testing, and User Evaluation.

3.1 Software Testing

The team decided to progress with a system of Test-After Development (TAD, sometimes called Test Later Development, or TLD), where code is written with code testing being conducted after the fact [4]. This development method, while not being the first choice of the team, should help to provide a better application overall due to some of the inherent benefits provided [6]:

- 1 **Better edge case tests** – As the testing is not done until after the code is written, the team will have a better understanding of edge cases relating to the features and as such will be able to write more comprehensive tests.
- 2 **Fewer trivial tests** – Tests will not be written with the sole goal of passing, meaning that less time will be wasted writing these inconsequential cases.
- 3 **Shorter timeline** – This is the greatest advantage of TAD over TDD, as testing after writing the code focuses the development team on the code that is being written without the distraction of switching between writing code and testing constantly.

3.1.1 Application flow

As the project took a test after development approach and was developed per feature, there were many issues with the flow of the application. Many basic elements had been neglected as the project neared the end, such as navigation from login to the game, and the game had no demo or explanation. There was confusion about where each part of the application was, and navigating was confusing for end users. As the scope of the application evolved throughout the project, the application had a slightly disjointed feel as the parts felt separate from each other and not unified. This was resolved when a clear aesthetic and conceptual direction was agreed on by the team.

3.1.2 Lack of Gamification

Sufficient gamification had not been achieved throughout the application for a long duration of the project. During initial testing phases users were not aware the application was a game. Many users did not find the game fun initially and said they would not play it again. Gamification theory was then studied by the team members and then built into the application. Concepts such as motivation and reward were included by adding a leaderboard, for example. Aesthetically, the app was transformed to be more game-like using characters for avatars and use of colours throughout. To add a narrative to the application, a how-to and explanation were created along with splash pages. The game was made highly configurable but also had pre-made games, to appeal to different users. It was noted by the team that when the leaderboard was first implemented, they became competitive and were encouraged to keep coming back to the application and making trades.

3.2 Accessibility Testing

Accessibility testing was carried out using a combination of automated and manual testing. For the automated testing the WebAIM Wave application was used [14], and for the manual test an expert in accessibility and an expert in usability jointly tested the final system. The automated testing suggested a number of elements that could be improved as regards to accessibility. It was highlighted that the navigation bar colour and navigation links had a low contrast. On the individual stock page, there is no alternative text for the news articles and for the Twitter profile photos. Furthermore, there are low contrasts between text and background colours for the Twitter hashtags and the sentiment badges. All of these issues were addressed as far as possible. The manual review process noted that the HTML code is “*well-structured and well-formed*” and advised that the essential components are all well labelled. The review was also noted that the site is accessible for people with a variety of visual and cognitive impairments. On the negative side, they noted that the text may have been a bit small in some specific places as well as the contrast between the background colour and white text potentially causing issues. All of these issues were addressed as far as possible.

3.3 User Testing

The User evaluation process was undertaken throughout the development of this application, and for convenience is presented below in the following stages:

- Phase 0: Market Research and Analysis
- Phase 1: Feature Prototyping and Design of Application
- Phase 2: End-to-End Usability Testing
- Phase 3: Autobiographical Testing
- Phase 4: End-to-End Usability Testing #2

★ Phase 0: Market Research and Analysis

This phase was undertaken as the project started, and the objective was to explore what people already know about the stock market and whether they would enjoy learning more about it by playing a game. The survey revealed that the respondents were interested in investing, and felt the need to do so, due to the cost-of-living crisis. Many also felt they did not know enough about the process, citing an inaccessibility of investment education. They also acknowledged games could be a useful tool to aid in their learning.

★ Phase 1: Feature Prototyping and Design of Application

This phase looked at how users reacted to various prototypes of the application using the 'Think Aloud' methodology (where the participants use the FinOptimise system, talking aloud as they move through the application or prototype and giving their thoughts) to assess their experiences. Extremely positive feedback was received, and a number of suggestions were given about the look-and-feel of the FinOptimise system.

★ Phase 2: End-to-End Usability Testing

Once an initial version of the FinOptimise system was completed, an analysis was undertaken of how usable the application is. This end-to-end usability testing is an extremely important phase for the user evaluation, and it was broken into two parts, which involved: (1) a cognitive walkthrough (where users are given specific tasks to do) by end users, most of whom had no investment experience, and (2) heuristic evaluations by expert reviewers coming from different technical backgrounds. Diversity in evaluators is the most effective way to find a large range of issues. Very few of the selected subjects had trading experience which was very helpful as they pointed out a lot of usability issues (and terminology issues) that would be resolved with an explanation. The expert evaluators all had a great technical proficiency, albeit in different areas, and they pointed out issues such as increasing the game-based features of the application (both in terms of format and content), as well as streamlining some of the features of the FinOptimise system.

★ Phase 3: Autobiographical Testing

Once the changes from the previous stage were completed, the development team themselves tested the FinOptimise system, focusing on three key considerations: (1) Flow of the application, (2) Gamification of the application, and (3) Functional testing. The system was evaluated on a number of hardware configurations, using both mobile and desktop devices. Many issues were resolved at this phase, including the flow between the game and the rest of the application, and ensuring the application was functioning consistently and reliably.

★ Phase 4: End-to-End Usability Testing #2

Based on the changes undertaken in all the previous phases, this phase involved testing the FinOptimise system with a combination of people who had already been involved in the test process, as well as new testers. The two main approaches to testing taken in this phase were: (1) User Evaluation using a cognitive walkthrough conducted with a group of 10 participants, and (2) a heuristic evaluation with expert evaluators.

Over the course of the evaluation process, the evaluators provided invaluable feedback to the development team, helping to guide the design of the application towards a more interactive and gamified experience. This evaluation feedback loop primarily took the form of UI design elements, two of which are detailed below.

Perhaps the UI design element that the evaluation process had the largest impact on was that of the Environmental, Social, Governance (ESG) ratings shown on the individual stock page. Initially, this was shown as a simple bar chart, however this was confusing to the users as the vertical scale was not consistent across all stocks on the application and as such provided meaningless information to the user. The next prototype for this was in the form of colourful sliders for each of the three ratings (along with one overall slider for the overall rating), this provided much clearer context for the user using pictograms representing the three categories. In evaluation, users much preferred this method of displaying ESG ratings as it very clearly showed on a spectrum where each company fell. Continuing on from this prototype, the development team had decided that it would be too complex to implement in the given timeline so it was instead simplified into its final form, which is a cleaner design using three vector symbols similar to a five star rating with the colour changing depending on the value of each.

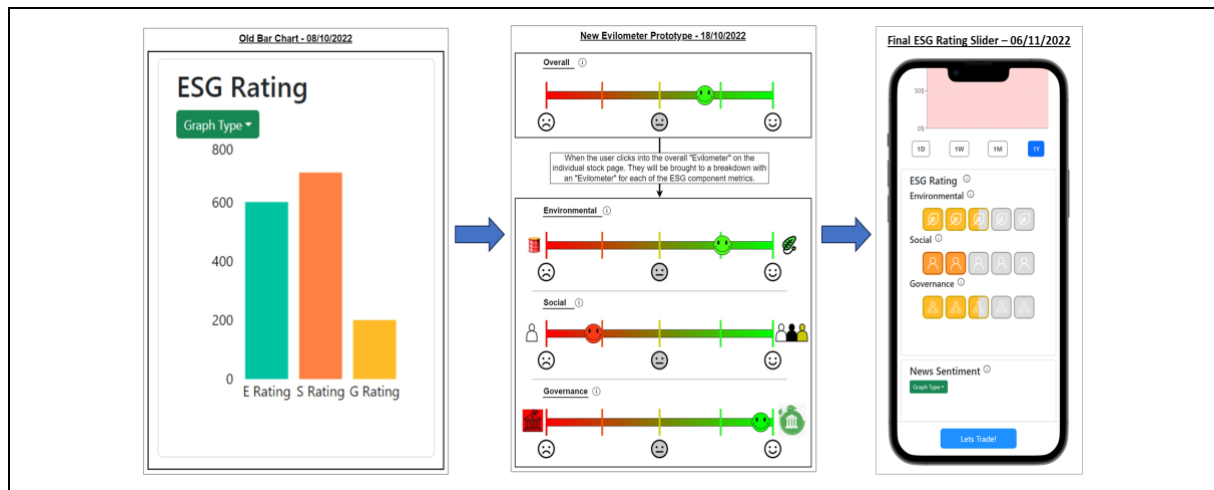


Fig. 3. Evolution of the ESG Rating in the FinOptimise system.

One of the most important UI elements to be refined in the evaluation process is that of the Stock Discovery screen. This is the screen that the users will see when they wish to explore stocks and as such it is the main gateway to exploring the content of the application. This went through many more prototypes/evaluations than the ESG ratings, usually with very minor changes from iteration to iteration. An example of 4 of the key iterations is shown below. The first image shows the initial prototype which was a vertical scrolling screen, when this was evaluated one user suggested the development team adopt a two-way scrolling screen similar to that of Netflix, with horizontal scrolling of grouped categories. Following on from this, the largest piece of feedback for the next two prototypes was that there was too much information shown on each. This resulted in the final design shown on the screenshot, which simplified the information shown to the user resulting in a cleaner and more engaging design.

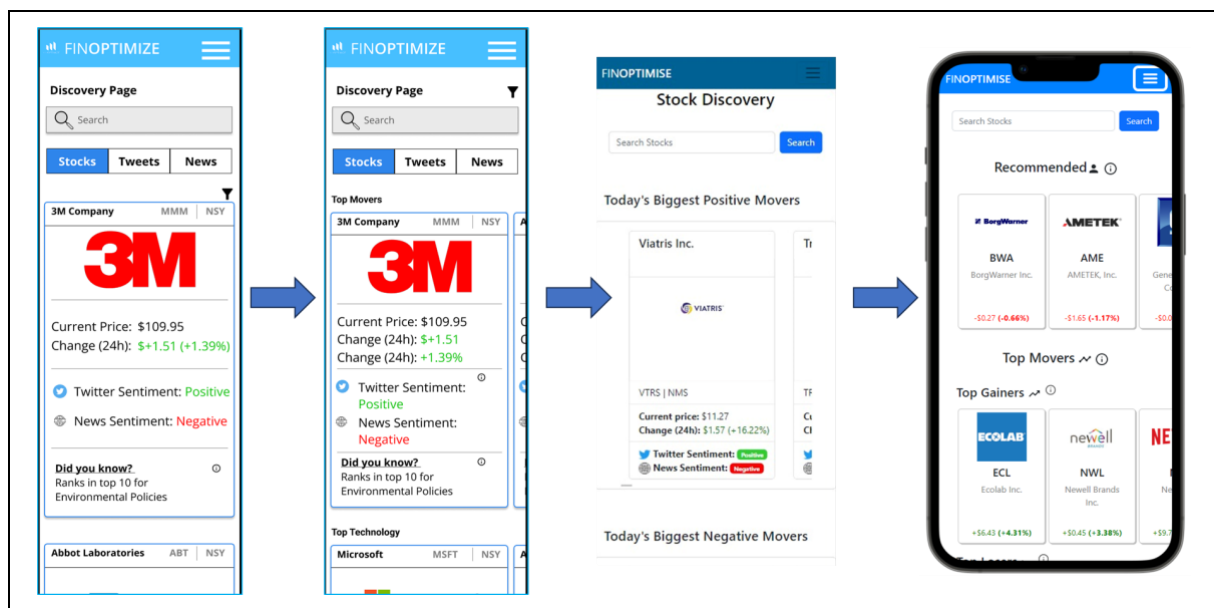


Fig. 4. Evolution of the Stock Discovery Page in the FinOptimise system.

A wide range of user feedback was collected as part of the evaluation process, and it is notable that after the FinOptimise system was completed, the user feedback was almost uniformly positive, and emphasised both the fun aspects of the system and the learning aspects of the system

One of the experts commented that the FinOptimise system was “*fun and enjoyable*” to play, and that they had “*learned an awful lot about stock investment, both the lingo and the techniques*”, from both the game and the website. They also commented that it seemed “*really similar to real stock apps*”.

4 CONCLUSIONS

The key reflections on this project are presented using the factors identified by Ravyse, *et al.* [13] as being important in a successful implementation of a serious game:

4.1 Game Narrative

The overall objective of the game is to help people learn about stock investment, and using a competition-based approach where players compete in a league proved to be an engaging and exciting way to help people learn about this topic. The associated website also helps support the game narrative, as it contains key terminology and a basic demo.

4.2 Realism

The final system is extremely similar to a real stock investment application, and, in fact, a number of real applications were reviewed during the design phase of this project to ensure that this was the case, so that the finished system would be as realistic as possible. Additionally all of the data used in the system is real-world data, including the stock prices, the media coverage, and the ESG ratings. Therefore, there would be little conceptual difficulty in moving from the FinOptimise system to a real investment application.

4.3 AI and Adaptivity

AI is used in the FinOptimise system both in the context of the recommender system and in the sentiment analysis of the media. The recommender system is an essential part of the system as it lets the players know which potential companies that they should invest in, The sentiment analysis provides up-to-date information media coverage of the companies that could be potentially invested in. The AI significantly increases the playability of the game.

4.4 Interaction

The interactive aspects of the system were considered some of the most important, where a great deal of user feedback and expert analysis were focused on the usability and interactivity of the system. A number of UX techniques were used throughout the project to ensure a high level of interaction, including the use of Personas, Wireframes, Nielsen's Heuristics, and Field studies

4.5 Feedback and Debrief

The FinOptimise provides the players with consistent feedback, updating them on stock prices, media commentary, and league table positions. Thus, the players learn within each game, and they can apply that learning in the next league. The website provides a feedback section where users can contribute questions and information to the system, as well as the "Learning" page to help players accelerate their learning.

4.6 Final Thoughts

Finally, the one factor that was not mentioned above, but that the development team deemed to be essential to the success of this project was the inclusion of users throughout the design and implementation process, their honest feedback, their guidance, and their suggestions turned this project into a much more fun, engaging, and educational system than it would have been without their help.

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